Ayush Kumar

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Portfolio | LeetCode: 200+ Solved | GeeksforGeeks | Bhubaneswar, Odisha, India

Summary

Final-Year B.Tech Information Technology student at IIIT Bhubaneswar with strong foundations in Data Structures & Algorithms, Full-Stack Software Engineering (Java, MERN, Python), and Applied Generative AI/LLM Systems. Experienced in architecting scalable RESTful APIs, autonomous agent workflows (LangGraph), vector search pipelines (pgvector), and containerized microservices (Docker, CI/CD). Actively seeking Full-Time Software Development Engineer (SDE) and Graduate Engineer roles.

Education

International Institute of Information Technology (IIIT) — **Bhubaneswar, Odisha**
Bachelor of Technology in Information Technology (CGPA: 7.48) — *Expected May 2026*

Delhi Public School, Nalco Nagar — **Angul, Odisha**
Class XII (Senior Secondary): 88.6% | Class X (Secondary): 98.2% — *2020 – 2022*

Technical Skills

Languages: Java (Core Java, OOP, Collections), JavaScript (ES6+), Python, TypeScript, SQL, C/C++, HTML5, CSS3
Web Technologies: React.js, Node.js, Express.js, FastAPI, Flask, RESTful APIs, Microservices, Vite, Tailwind CSS, Bootstrap 5
AI, Machine Learning & GenAI: LangGraph, Autonomous Agents, RAG Pipelines, pgvector, Scikit-learn, XGBoost, Pandas, NumPy, Google Gemini API
Databases & DevOps: PostgreSQL 16, MongoDB Atlas, Docker, Docker Compose, Git, GitHub Actions (CI/CD), Linux, Render, Cloudinary
Core CS Fundamentals: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, System Design

Projects

AgentIQ | *Python, FastAPI, LangGraph, PostgreSQL, pgvector, React, TypeScript, Docker Compose* GitHub | Live Demo
- Architected an enterprise agentic RAG platform using **FastAPI** and **LangGraph**, orchestrating bounded multi-step reasoning (≤ 4 steps) with deterministic fallbacks, execution timeouts, and sub-second retrieval latency.
- Implemented hybrid semantic search via **PostgreSQL 16** and **pgvector** (HNSW indexing, 0.45 cosine cutoff) with source citations, protected by a 5-layer SQL guardrail with **sqlglot AST parsing** and RBAC.
- Constructed an enterprise-grade 3-pane UI in **React 18** and **TypeScript**, containerizing full-stack services with multi-stage **Docker** and establishing automated **GitHub Actions CI/CD** testing.

Feasto | *React.js, Node.js, Express.js, MongoDB Atlas, Razorpay, Google Gemini API, Docker* GitHub | Live Demo
- Engineered a production-grade food delivery ecosystem following **MERN architecture** with decoupled customer and admin management portals, featuring real-time driver delivery tracking with **Leaflet GPS** and persistent cart state.
- Integrated **Razorpay Payment Gateway** with cryptographic **HMAC-SHA256** webhook signature verification, ensuring zero transaction discrepancies across shopping carts, order lifecycles, and database states.
- Embedded an AI food concierge using **Google Gemini API** for personalized dish recommendations; implemented JWT role-based access control (RBAC), Cloudinary CDN image pipelines, and multi-container **Docker** deployment.

StayNest | *Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5, Cloudinary, Razorpay, PDFKit* GitHub | Live Demo
- Developed a full-stack accommodation marketplace adopting **MVC architecture** with full CRUD listings, geospatial filters, dynamic reviews, and responsive UI built with Bootstrap 5.
- Engineered a collision-proof booking engine featuring interactive blackout calendar scheduling, Razorpay payment verification, and automated server-side PDF invoice generation using **PDFKit**.
- Hardened backend application security with **Passport.js** session authentication, Joi input validation schemas, Helmet HTTP security headers, and MongoDB Atlas cloud storage.

Smart House Price Predictor | *Python, Flask, XGBoost, Scikit-learn, Pandas, NumPy, Docker* GitHub | Live Demo
- Developed a California real estate valuation engine achieving $R^2 = 83.93\%$ and 17.52% **MAPE** ($30,245 MAE) using an optimized **XGBoost** gradient-boosted ensemble trained on 20,000+ property records.
- Integrated distribution-free **Split Conformal Prediction** delivering mathematically guaranteed 90% **confidence intervals** (±$68,169), eliminating financial tail risk compared to traditional point estimates.
- Implemented Explainable AI (**XAI**) feature attribution telemetry and defensive input validation, deploying via a low-latency (<20ms) **Flask REST API** containerized with Docker and automated **GitHub Actions CI**.

Achievements & Coding Profiles

- Solved **200+ Data Structures & Algorithms** problems in **Java** across **LeetCode** and **GeeksforGeeks**, demonstrating algorithmic optimization and problem-solving skills.
- Architected, containerized, and deployed **4 production systems** (AgentIQ, Feasto, StayNest, SmartHouse) with Docker, CI/CD pipelines, and cloud hosting on Render.
- Demonstrated sustained academic excellence with **98.2% in Class X** and **88.6% in Class XII**; Final-Year B.Tech IT student at **IIIT Bhubaneswar**.